Sarah A. Zhao

saqzhao@gmail.com • linkedin.com/in/sarah-a-zhao • saqzhao.github.io • ORCID 0009-0008-6068-5693

EDUCATION

The Scripps Research Institute, La Jolla, California, USA
Student, Ph.D. in Chemical and Biological Sciences *Aug 2026—Present*
University of Oxford, Oxford, England
Student, D.Phil. in Biochemistry *Aug 2026—Present*
Massachusetts Institute of Technology, Cambridge, Massachusetts
Master of Engineering, Computer Science (BioEECS) | GPA: 4.5/5.0 *May 2024—May 2025*

- Advised by Prof. Manolis Kellis, Ph.D.
- Thesis: Modeling Sequence Uncertainty in Comparative Genomics with a Probabilistic DNA Representation
- Relevant Coursework: *Genetics, Mining Microbiomes, Synthetic Biology, Biochemistry, *Biomechanics

Bachelor of Science, Computer Science & Engineering, Mathematics | GPA: 4.3/5.0 *Sept 2020—May 2024*

- Relevant Coursework: Organic Chemistry, Biology, Differential Equations, Linear Algebra, *Machine Learning, *Natural Language Processing, *Computer Vision, Advanced Algorithms (* denotes graduate level)

AWARDS

MIT David Adler Memorial Thesis Award *May 2026*

- Outstanding MEng thesis in the MIT EECS department, for research done in the Kellis Lab

Fulbright Fellowship *May 2025*

- Germany open-study/research student award
- Selected by the US Department of State to conduct funded research abroad, in the Typas lab at EMBL

MIT Anna Pogosyants UROP Award *May 2025*

- Outstanding undergraduate research in the MIT EECS department for developing pDNA in the Kellis lab

Skaggs-Oxford Scholarship *April 2025*

- Full funding for five years of doctoral study at the Scripps Research Institute and the University of Oxford

EXPERIENCE

Scripps Research Institute, La Jolla, California, USA *Aug 2026—Present*
PhD Student
Fulbright Research Fellow, Laboratory of Athanasios Typas, Ph.D.

- Rotation with Prof. Megan Ken on dimerization initiation site of HIV 5' untranslated region

University of Oxford, Oxford, England *Aug 2026—Present*
Postgraduate Student
European Molecular Biology Laboratory, Heidelberg, Germany *Sept 2025—June 2026*
Fulbright Research Fellow, Laboratory of Athanasios Typas, Ph.D.

- Mapping protein-protein interaction networks within *Bacteroides Uniformis* and *Phocaeicola vulgatus* to develop as model species for microbiome research using AlphaFold3
- High-throughput transposon library screening of polysaccharide utilisation loci in *B. Uniformis* and *P. vulgatus*

Broad Institute of MIT and Harvard, Cambridge, Massachusetts, USA
Graduate Student, Laboratory of Manolis Kellis, Ph.D. *Oct 2024—May 2025*
Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Master's Student, Laboratory of Manolis Kellis, Ph.D. *Sept 2024—May 2025*

- Variant interpretation for precision medicine
- Built a pipeline to predict the effect of genetic variation on the dysregulation of molecular components in psychiatric disorders

Undergraduate Researcher, Laboratory of Manolis Kellis, Ph.D. *Jan 2024—Aug 2024*

- Direct Advisor: Riley Mangan, Ph.D.
- Developed infrastructure using Go for probabilistic representations of biological sequence data, contributing to open-source genomics software package Gonomics
- Reconstructed ancestral genome from extant species genomes based on posterior predictive distributions to identify derived gene regulatory changes between ancient and modern genomes

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Undergraduate Researcher, Laboratory of Lindsay Case, Ph.D. *Feb 2024—Aug 2024*

- Direct Advisor: Jibin Sadasivan, Ph.D.
- Constructed protein-protein interaction network using Python and Cytoscape to identify potential scaffold proteins that phase separate in focal adhesions
- Designed and conducted experiments to investigate co-localization between candidate phase separating proteins and known proteins in focal adhesions
- Conducted comprehensive literature review on the focal adhesion proteome and phase separation

MIT Mathematics Directed Reading Program, Cambridge, Massachusetts, USA *Jan 2024*

- Studied and presented on parameterized algorithms (based on work by Cygan et al., 2015)

Kongsberg Maritime, Lysaker, Norway *June 2023—Aug 2023*
Machine Learning Research Intern
- Developed machine-learning video compression algorithms (PyTorch Lightning) to decrease file size and improve latency

NCSoft, Seongnam, South Korea *June 2022—Aug 2022*
Machine Learning Research Intern
- Built a text-to-video model (Pytorch) generating lifelike text-to-animation with multilingual input
- Designed morph targets in Maya for Mandarin lip synching to localise Korean content for Taiwan
- Refactored codebase to comply with current industry code standards and streamline development

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Undergraduate Researcher, Laboratory of Faez Ahmed, Ph.D. *Feb 2022—May 2022*
- Designed algorithm (Python) to process publication abstracts and metadata using natural language programming (NLTK, scikit-learn)
- Evaluated data trends based on statistical data analysis & visualised trends (Plotly, Dash)

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Undergraduate Researcher, Laboratory of Deblina Sarkar, Ph.D. *Oct 2020—Sept 2021*
- Programmed models in Python to analyse circuit impedance and fabricated microcoils for non-invasive nanosensors monitoring cell response
- Conducted literature review on neural sensor material impedance, conductivity, and thermal activity

PUBLICATIONS

- Yoo D, … **Zhao SA**, … Eichler, EE. (2025). Complete sequencing of ape genomes. *Nature*, 1–18. https://doi.org/10.1038/s41586-025-08816-3

(In Preparation)

- Luo Y, … **Zhao SA**, … Lowe CB. (Submitted 2025, Cell Genomics) Intraspecific sequence variation and complete genomes refine the identification of rapidly evolved regions in humans. *bioRxiv*. https://doi.org/10.1101/2025.10.20.683446
- **Zhao SA**, …, Lowe CB, Kellis M. (In preparation) pDNA: A High-Performance Software Package for Probabilistic Representations of Nucleotide Sequences.

TEACHING

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Teaching Assistant, 6.042/6.1200 Mathematics for Computer Science *Sept 2022—May 2024*
- Led twice-weekly recitations for ~15 students covering discrete maths and computer science for four semesters
- Developed problem sets and exam questions
- Held office hours weekly (4 hrs) for >300 student class
- Supervised course graders (~20 undergraduate graders)

Lab Assistant, 6.009/6.1010 Fundamentals of Programming *Feb 2022—May 2022*
- Advised students in one-on-one meetings during office hours and weekly oral lab presentations
- Graded student labs and homework assignments

Grader, 8.01 Physics I: Classical Mechanics *Feb 2022—May 2022*
- Evaluated student work and provided feedback

TECHNICAL SKILLS

- **Programming Languages:** *Advanced proficiency in* Python, Go, Typescript, R, bash; *Proficient in* Java, C, Assembly; *Additional experience in* Julia, Matlab, Javascript, HTML and CSS
- **Machine Learning, Statistics, and Data Science:** PyTorch, Tensorflow, NumPy, Pandas, SKLearn
- **Experimental**: molecular cloning, DNA/RNA extraction, gel electrophoresis, cell culture, western blotting, immunoprecipitation,, high-throughput liquid handling and robotics, microfluidics, anaerobic culture
- **Languages:** Mandarin (Advanced Proficiency), Spanish (Proficient), Korean (Conversational)

COMMUNITY INVOLVEMENT

MIT 2024 Ring Committee, Cambridge, Massachusetts, USA
Contributing Artist, Social Chair, *June 2021—June 2022*
- Planned events for >1200 students, managed correspondence, collaborated with jeweller representatives

MIT ActLingual, Cambridge, Massachusetts, USA
Committee Leader *Sept 2020—May 2022*
- Wrote medical interpreting manual for Spanish and Mandarin, & facilitated student interpretation opportunities

SERVICE

MIT MedLinks, Cambridge, Massachusetts, USA
Residential Director *Sept 2022—May 2024*

- Connected students to healthcare services, ran education campaigns, planned welfare events for ~200 students

Harvard Phillips Brooks House Association Chinatown Citizenship, Cambridge, Massachusetts, USA
Mock Interview Teacher *Sept 2020—May 2022*

- Facilitated mock interviews and designed curriculum for Chinese-native students for U.S. naturalisation exam

MIT ARCTAN (Red Cross), Cambridge, Massachusetts, USA
Community Service Coordinator *Sept 2020—May 2022*

- Coordinated events such as education campaigns, blood drives, CPR training and volunteering